

Math728J – Midterm Exam

1. Suppose a matrix $\mathbf{A} \in \mathbb{R}^{N \times N}$ is an M-matrix, prove the following results:
 - a. $\mathbf{A} + s\mathbf{I}$ is an M-matrix for any $s > 0$.
 - b. there exists a matrix $\mathbf{B}$ with $\mathbf{B} \geq 0$ and a number $s > \rho(\mathbf{B})$ such that $\mathbf{A} = s\mathbf{I} - \mathbf{B}$ holds.
2. Given a spd matrix $\mathbf{A} \in \mathbb{R}^{N \times N}$ and a vector $\mathbf{b}$. Show that if $\mathbf{A}$ has constant diagonal then PCG with Jacobi preconditioning produces the same iterates as CG with no preconditioning.
3. Consider a matrix of form $\mathbf{A} = \mathbf{I} + \alpha\mathbf{B}$ where $\mathbf{B}$ is real skew symmetric (i.e., $\mathbf{B} = -\mathbf{B}^T$).
 - a. show that $(\mathbf{A}\mathbf{x}, \mathbf{x})/(\mathbf{x}, \mathbf{x}) = 1$ for all nonzero $\mathbf{x}$.
 - b. show that the resulting Hessenberg matrix by the Arnoldi process will have the following tridiagonal form:

$$\mathbf{H}_k = \begin{pmatrix} 1 & -\eta_2 & & & \\ \eta_2 & 1 & -\eta_3 & & \\ & & \ddots & & \\ & & & \eta_{k-1} & 1 & -\eta_k \\ & & & & \eta_k & 1 \\ & & & & & \eta_{k+1} \end{pmatrix}.$$

4. Let

$$\mathbf{A} = \begin{pmatrix} 0 & & & 1 \\ 1 & 0 & & \\ & \ddots & \ddots & \\ & & 1 & 0 \end{pmatrix} \in \mathbb{R}^{N \times N}, \quad \mathbf{b} = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad \mathbf{x}^* = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix}.$$

Clearly $\mathbf{x}^*$ is the exact solution of the linear system $\mathbf{A}\mathbf{x} = \mathbf{b}$.

- a. show how GMRES will converge on the linear system $\mathbf{A}\mathbf{x} = \mathbf{b}$ with $\mathbf{x}_0 = 0$.
 - b. show how CGNR will converge on the linear system $\mathbf{A}\mathbf{x} = \mathbf{b}$ with $\mathbf{x}_0 = 0$.
5. Let $\mathbf{A} \in \mathbb{R}^{N \times N}$. Assume that the Lanczos process for generating a biorthogonal system $\mathbf{V}_{k+1}$ and $\mathbf{W}_{k+1}$ does not break down before step k .
 - a. show that $\mathbf{V}_{k+1}$ and $\mathbf{W}_{k+1}$ are both of full rank.
 - b. prove the following identities:

$$\mathbf{A}\mathbf{V}_k = \mathbf{V}_k\mathbf{T}_k + \delta_{k+1}\mathbf{v}_{k+1}\mathbf{e}_k^T, \quad \mathbf{A}^T\mathbf{W}_k = \mathbf{W}_k\mathbf{T}_k^T + \beta_{k+1}\mathbf{w}_{k+1}\mathbf{e}_k^T, \quad \mathbf{W}_k^T\mathbf{A}\mathbf{V}_k = \mathbf{T}_k$$